THE UNIVERSITY OF MELBOURNE

| Student number |
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Assignment 7 Semester 1, 2023

School of Mathematics and Statistics

# MAST10012 Introduction to Mathematics

Submission deadline: 5:00 PM (17:00, AEST), Thursday 4 May 2023

This assignment consists of 5 pages (including this page)

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## Instructions to Students

*Writing*

- If you have a printer (or can access one), print the assignment one-sided. Write your answers in the boxes provided on the assignment that you have printed.
- If you do not have a printer, and you know how to annotate a PDF using a tablet, then annotate your answers directly onto the assignment PDF and save a copy for submission.
- If you cannot use the template, write your answers on blank A4 paper. Mimic the template by numbering your pages, and note which answers to write on each page.
- Record your student ID number in the top right corner of this cover page (or the first page of your submission), and your details at the bottom of this page (or the first page).
- Provide complete and fully justified solutions for each question.
- Write legibly, and use appropriate notation. Marks will not be awarded to unclear work.
- If you run out of space for a question, append additional work to the end of your submission. Make a note ("See page X") in the relevant question box warning the marker.
- Teaching staff cannot help you directly with assignment questions. Please ask on the Ed Discussion board if you need clarification on the wording of a question.

*Scanning*

- Put the pages in number order and the correct way up. Add any extra pages to the end. Use a scanning app to scan all pages to PDF. Do not scan on an angle. Crop images to the page borders. Make sure that your PDF file is readable. Material that cannot be read will not be marked.

*Submitting*

- Go to the Gradescope window. Choose the relevant Canvas assignment. Make sure that you upload the correct PDF file. You should get a Gradescope confirmation email.
- Assignments submitted late will incur a penalty of 4% (of the total mark available) for every hour or part thereof up to 24 hours. Written assignments cannot be submitted more than one day (24 hours) late.

*Student Details*

- Student name: _____
- Tutorial A tutor name: _____
- Tutorial A day and time (e.g. Monday 9AM): _____

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**Question 1**

Follow the listed steps to sketch this function:

$$f(x) = (3-x)^2(2x-1)^3$$

(a) Find the maximal domain of $f$ and any points of discontinuity.

(b) Find the $y$-intercept(s).

(c) Find the $x$-intercept(s) (root or roots).

(d) Find the sign of the function over notable intervals (either side of roots and points of discontinuities).

(e) Find any horizontal and vertical asymptotes.

(f) Differentiate $f(x)$. You do not need to simplify, expand or factorise the result.

(g) Find the critical points. Note that the derivative can be factorised to $f'(x) = 10(2x-1)^2(x^2-5x+6)$.

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(h) Partition the domain according to the critical points and classify each interval according to whether the function increases or decreases.

(i) Evaluate the function at each of the critical points and classify the critical points.

(j) Combine the information from previous parts using a number line for $f(x)$ and a number line for $f'(x)$.

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(k) Sketch the function $f(x)$. Your sketch must be large, hand-drawn (not digitally produced), legible, and clearly label all key features.

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**End of Assignment**